



Cs301 Quiz 16 August 2021

Suppose there is an image segmented into pixels. Each pixel has _____ neighbour(s).

Select the correct option

- ☐ 0
- ☒ 4
- ☐ 8
- ☐ 16

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The array in binary search is sub divided _____.

Select the correct option



Once



Twice

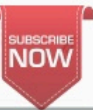


N time



Until a sublist is no more divisible

AM Knowledge Official



If a hash function returns 4 for more than one value, this condition is called _____.

Select the correct option



Merging



Collision

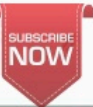


Mapping



Clustering

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Binary Search can be categorized into which of the following?

Select the correct option



Greedy algorithm



Dynamic programming

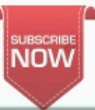


Divide and conquer



Brute Force technique

AM Knowledge Official



A table consist of several columns, known as

Select the correct option

☒	Fields	
☐	Entity	AM Knowledge Official
☐	Tuple	
☐	Record	

If unions are done by weight (size), the depth of any element is never greater than

Select the correct option



$$\log_2 n$$



$$n \log_2 n$$



$$\log 3n$$



$$\log n^n$$

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Which of the following is NOT an open addressing technique to resolve collisions ?

Select the correct option



Quadratic probing



Double hashing



Cubic probing



Linear probing

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What is a skip list?

Select the correct option

- ☐ a linked-list with size value in nodes
- ☐ a tree which is in the form of linked list
- ☒ a linked-list that allows faster search within an ordered sequence
- ☐ a linked-list that allows slower search within an ordered sequence

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CS301

Quiz # 02

02

Suppose there is an image of $7 * 7$. now we will have matrix of _____ rows and _____ columns.

Select the correct option

- | | | |
|----------------------------------|---------|------------------------------|
| ☒ | 7.7 | |
| ☐ | 8.8 | <i>AM Knowledge Official</i> |
| ☐ | 49.49 | |
| ☐ | 100.100 | |

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Consider a hash table of size seven, with starting index zero, and a hash function $(3x + 4) \bmod 7$. Assuming the hash table is initially empty, which of the following is the contents of the table when the sequence 1, 3, 8, 10 is inserted into the table using closed hashing? Note that '_' denotes an empty location in the table.

Select the correct option

- | | | |
|----------------------------------|----------------------|------------------------------|
| ☐ | 8, _, _, _, _, _, 10 | |
| ☐ | 1, _, _, _, _, _, 3 | <i>AM Knowledge Official</i> |
| ☐ | 1, 10, 8, _, _, _, 3 | |
| ☒ | 1, 8, 10, _, _, _, 3 | |

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Question # 9 of 10 (Start time: 02:31:46 AM, 16 August 2021)

Total Marks: 1

In 1990,Bill Pugh proposed an enhancement on linked lists and the new data structure was termed as

Select the correct option

- | | |
|----------------------------------|----------------------|
| ☐ | Tree List |
| ☐ | Enhanced Linked list |
| ☒ | Skip list |
| ☐ | Fixed list |



Question # 8 of 10 (Start time: 02:31:03 AM, 16 August 2021)

Total Marks: 1

The array in binary search is sub divided _____.

Select the correct option

- ☐ N time
- ☐ Once
- ☒ Until a sublist is no more divisible
- ☐ Twice



Question # 7 of 10 (Start time: 02:29:58 AM, 16 August 2021)

Total Marks: 1

Which of the following is NOT true regarding the maze generation?

Select the correct option

- ☐ Removing a wall is the same as doing a union operation.
- ☐ Do not remove a randomly chosen wall if the cells it separates are already in the same set.
- ☒ Remove a randomly chosen wall if the cells it separates are already in the same set.
- ☐ Randomly remove walls until the entrance and exit cells are in the same set.



Question # 6 of 10 (Start time: 02:28:59 AM, 16 August 2021)

Total Marks: 1

During union by size method, which data structure is used to improve the balancing of tree?

Select the correct option

- | | |
|----------------------------------|-------------|
| ☒ | Array |
| ☐ | Tree |
| ☐ | Stack |
| ☐ | Linked List |



Question # 5 of 10 (Start time: 02:28:29 AM, 16 August 2021)

Total Marks: 1

What is the time complexity of binary search with iteration?

Select the correct option

- | | |
|----------------------------------|---------------|
| ☐ | $O(n)$ |
| ☐ | $O(n \log n)$ |
| ☐ | $O(n^2)$ |
| ☒ | $O(\log n)$ |



Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 02:28:05 AM, 16 August 2021)

Total Marks: 1

Binary Search can be categorized into which of the following?

Select the correct option

- ☐ Brute Force technique
- ☐ Dynamic programming
- ☐ Greedy algorithm
- ☒ Divide and conquer



Question # 3 of 10 (Start time: 02:27:39 AM, 16 August 2021)

Total Marks: 1

Heap can be used to implement

Select the correct option

- | | |
|----------------------------------|----------------|
| ☐ | Linked list |
| ☐ | Stack |
| ☒ | Priority Queue |
| ☐ | Queue |



Question # 2 of 10 (Start time: 02:27:11 AM, 16 August 2021)

Total Marks: 1

A binary relation R over S is called an equivalence relation if it has following property(s)

Select the correct option

- | | |
|----------------------------------|--------------------------|
| ☒ | All of the given options |
| ☐ | Reflexivity |
| ☐ | Transitivity |
| ☐ | Symmetry |



Question # 1 of 10 (Start time: 02:26:07 AM, 16 August 2021)

Total Marks: 1

During the union by size method, all the array element initialized to -1 shows;

Select the correct option

- ☐ Every tree has two elements each.
- ☒ None of given.
- ☐ Tree is complete now
- ☐ Initial condition of tree



Question # 10 of 10 (Start time: 07:55:20 AM, 16 August 2021)

Total Marks

Consider a hash table of size seven, with starting index zero, and a hash function $(3x + 4) \bmod 7$. Assuming the hash table is initially empty, which of the following is the contents of the table when the sequence 1, 3, 8, 10 is inserted into the table using closed hashing? Note that '_' denotes an empty location in the table.

Select the correct option

- | | |
|----------------------------------|-------------------|
| ☐ | 1, _, _, _, _, 3 |
| ☐ | 8, _, _, _, _, 10 |
| ☒ | 1, 8, 10, _, _, 3 |
| ☐ | 1, 10, 8, _, _, 3 |

Question # 9 of 10 (Start time: 07:54:37 AM, 16 August 2021)

In 1990,Bill Pugh proposed an enhancement on linked lists and the new data structure was termed as

Select the correct option



Skip list



Fixed list



Tree List



Enhanced Linked list

Suppose a hash function returns 6 for the given value. At which index of array the value will be saved?

Select the correct option



5



7



0



6

A binary relation R over S is called an equivalence relation if it has following property(s)

Select the correct option

- | | |
|----------------------------------|--------------------------|
| ☐ | Reflexivity |
| ☒ | All of the given options |
| ☐ | Symmetry |
| ☐ | Transitivity |

Question # 6 of 10 (Start time: 07:51:53 AM, 16 August 2021)

Total Marks

Which of the following possible operations are performed on Table ADT?

Select the correct option

- | | |
|----------------------------------|------------------------|
| ☐ | Only Insert and Find |
| ☒ | Insert, Find, Remove |
| ☐ | Only Insert and Remove |
| ☐ | Only Find and Remove |

Binary Search can be categorized into which of the following?

Select the correct option

- | | |
|----------------------------------|-----------------------|
| ☒ | Divide and conquer |
| ☐ | Dynamic programming |
| ☐ | Greedy algorithm |
| ☐ | Brute Force technique |

Which of the following is true regarding the maze generation?

Select the correct option

- ☐ Do not remove a randomly chosen wall if the cells it separates are already in the same set
- ☐ Randomly remove walls until the entrance and exit cells are in the same set
- ☐ Removing a wall is the same as doing a union operation
- ☒ All of the given

Question # 3 of 10 (Start time: 07:47:49 AM, 16 August 2021)

Total Marks: 10

If unions are done by weight (size), the depth of any element is never greater than

Select the correct option

- | | |
|----------------------------------|----------------------|
| ☐ | $\log 3n$ |
| ☒ | $\log \frac{n}{2}$ |
| ☐ | $n \log \frac{n}{2}$ |
| ☐ | $\log n^n$ |



Suppose there are 100 elements in an equivalence class, so initially there will be 100 trees. The collection of these trees is called _____.

Select the correct option

- | | |
|----------------------------------|---------|
| ☒ | Forest |
| ☐ | Class |
| ☐ | Cluster |
| ☐ | Bunch |

Question # 1 of 10 (Start time: 07:44:28 AM, 16 August 2021)

Total Marks: 1

Given the values are the array representation of heap:

12 23 26 31 34 44 56 64 78 100

What is the 5th smallest element in the given heap?

Select the correct option



44



34



56



31

Question # 10 of 10 (Start time: 02:11:46 PM, 16 August 2021)

Total Marks: 1

What is the time complexity of binary search with iteration?

Select the correct option

☐	$O(n^2)$
☒	$O(\log n)$
☐	$O(n)$
☐	$O(n \log n)$

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2021 || MTH621 100% SOLVED QUIZ

QUIZ NO 3
SPRING 2021
100%
ACCURATE ANSWER

3:42

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Question # 7 of 10 (Start time: 02:10:23 PM, 16 August 2021)

Total Marks: 1

A hash function returns a _____ value.

Select the correct option

☐	Float
☒	Integer
☐	Char
☐	Double

[Click to Save Answer & Move to Next Question](#)

Question # 6 of 10 (Start time: 02:09:51 PM, 16 August 2021)

Total Marks: 1

Given the values are the array representation of heap;
12 23 26 31 34 44 56 64 78 100
What is the 5th smallest element in the given heap?

Select the correct option

☒	34
☐	56
☐	44
☐	31

[Click to Save Answer & Move to Next Question](#)

Question # 4 of 10 (Start time: 02:08:41 PM, 16 August 2021)

Total Marks: 1

If unions are done by weight (size), the depth of any element is never greater than

Select the correct option

- | | | |
|----------------------------------|-----------------|----|
| ☐ | $\log 3n$ | // |
| ☒ | $\log n.$
2 | // |
| ☐ | $n \log n$
2 | // |
| ☐ | $\log n*n$ | // |

[Click to Save Answer & Move to Next Question](#)

Question # 1 of 10 (Start time: 02:06:44 PM, 16 August 2021)

Total Marks: 1

Suppose there is an image segmented into pixels. Each pixel has _____ neighbour(s).

Select the correct option

☒	4	//
☐	16	//
☐	8	//
☐	0	//



Click to Save Answer & Move to Next Question

Question # 2 of 10 (Start time: 02:08:01 PM, 16 August 2021)

Total Marks: 1

The union operation is based on size or weight but the reducing the in-between links or path compression from ____ to the ____ is done by the find method.

Select the correct option

- | | | |
|----------------------------------|--------------|----|
| ☐ | node , node | // |
| ☒ | root , root | // |
| ☐ | nodes , root | // |
| ☐ | roots , node | // |

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 01:43:54 PM, 16 August 2021)

Total Marks: 1

A table consists of several columns, known as

Select the correct option

☐	Entity
☐	Tuple
☒	Fields
☐	Record

[Click to Save Answer & Move to Next Question](#)

Question # 3 of 10 (Start time: 01:36:40 PM, 16 August 2021)

Total Marks: 1

The union operation is based on size or weight but the reducing the in-between links or path compression from ____ to the ____ is done by the find method.

Select the correct option

- | | |
|----------------------------------|--------------|
| ☒ | nodes , root |
| ☐ | roots , node |
| ☐ | node , node |
| ☐ | root , root |

[Click to Save Answer & Move to Next Question](#)

Question # 7 of 10 (Start time: 01:41:47 PM, 16 August 2021)

Total Marks: 1

Suppose there is an image of 7×7 . now we will have matrix of ____ rows and ____ columns.

Select the correct option

- | | |
|----------------------------------|----------|
| ☐ | 100, 100 |
| ☐ | 49, 49 |
| ☐ | 8, 8 |
| ☒ | 7, 7 |

[Click to Save Answer & Move to Next Question](#)

Question # 8 of 10 (Start time: 01:42:36 PM, 16 August 2021)

Total Marks: 1

Which of the following is NOT true regarding the maze generation?

Select the correct option

- | | |
|----------------------------------|---|
| ☐ | Randomly remove walls until the entrance and exit cells are in the same set. |
| ☐ | Do not remove a randomly chosen wall if the cells it separates are already in the same set. |
| ☒ | Remove a randomly chosen wall if the cells it separates are already in the same set. |
| ☐ | Removing a wall is the same as doing a union operation. |

[Click to Save Answer & Move to Next Question](#)

Question # 2 of 10 (Start time: 01:35:04 PM, 16 August 2021)

Total Marks: 1

Which of the following is true regarding the maze generation?

Original

Select the correct option

- | | |
|----------------------------------|--|
| ☐ | Randomly remove walls until the entrance and exit cells are in the same set |
| ☐ | Removing a wall is the same as doing a union operation |
| ☒ | All of the given |
| ☐ | Do not remove a randomly chosen wall if the cells it separates are already in the same set |

Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 01:33:10 PM, 16 August 2021)

Total Marks: 1

The expression
if (! heap->isEmpty())
checks

Select the correct option

- | | | |
|----------------------------------|------------------------|----|
| ☐ | Heap is empty | // |
| ☐ | Heap is full | // |
| ☒ | Heap is not empty | // |
| ☐ | Not a valid expression | // |



Click to Save Answer & Move to Next Question